

2025 AMC 10B Problem 4 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 4. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 4



Problem:

The value of the two-digit number $\underline{a}\underline{b}$ in base seven equals the value of the two-digit number $\underline{b}\underline{a}$ in base nine. What is $a + b$?

Answer Choices:

- A. 7
- B. 9
- C. 10
- D. 11
- E. 14



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Solution:

The base-7 number $\overline{AB}_7$ equals the base-9 number $\overline{BA}_9$:

$$7a + b = 9b + a.$$

Thus

$$7a + b = 9b + a \implies 6a = 8b \implies 3a = 4b.$$

So $a = 4k, b = 3k$ for some integer k . Since a, b are digits in base 7, we require $1 \leq a, b \leq 6$. The only possibility is $k = 1$, giving $(a, b) = (4, 3)$.

Hence

$$a + b = 4 + 3 = \boxed{\text{(A) } 7}.$$

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